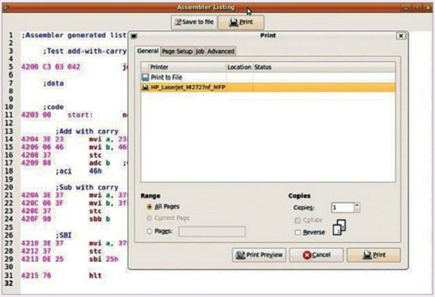

Fig. 2: The Assembler listing of GNUSim

machine code. Major elements of Assembler include Mnemonics, which are instruction strings of operations; Labels, which are targeted named points in the code for Call commands; Comments, which are not part of the code but are remarks for the coder's convenience; and Pseudo-opcodes.

There are three sub-menus in the Assembler menu. The first is Assemble, which loads the program code to the memory address. The second is Execute, which compiles and runs the loaded program. The third option is Show Listing, where users can view the executed program code along with the relevant memory address opcode, mnemonics and comments. The main importance of this option lies in preparing the program for moving into the actual hardware by either saving as a file and printing or viewing the complete listing.

Debug and reset. The simulator toolbar has two other menus. One is Debug. It consists of sub-menus Step In, Step Over and Step Out. These make code debugging easy by analysing the register and memory content in each step. Debug menu also comes with a code breakpoint feature.

Reset menu can help users to erase old data from the simulator and reset values of registers, ports, flags and memory. Reset can be done individually for each component or for all at once.

Updates in v1.3.7

The latest version of GNUSim8085 has a strong debugger unit. The program automatically stops running in case of infinite-loop errors or any program execution fault. Stack tracing is quite efficient in this version. For instance, users are notified in case more Pop commands are executed than Push.

Moreover, the latest version saw fixing of some major bugs in the previous releases. One among them is the unavailability of Project Save prompt when switching to a new project. Frequent crashing of the software when clicking Help→About has also been solved. Another major bug solved was the flag setting error of DAA.

Finally, the biggest advantage of the software is enabling developers to insert the program through the Intel 8085 mnemonics and avoid hand assembly. Through this, developers can save a great deal of time in debugging and ensure an error-free programming of the hardware module. EFY